

DPCN Assignment 2

Members

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Title: Network Analysis of EEG Data

Abstract

- The aim of the analysis is to reveal causal relationships between different brain regions and their temporal dynamics.
- The dataset contains resting state EEG data of 103 infants.
- Application of statistical tests on this network gives us insight into the network dynamics and how the brain develops in the first year of the infant's life.

Introduction

Current Literature

- **EEG Functional Connectivity Across Infancy (2024, PMC11314780)**

This study analyzed EEG functional connectivity in 146 infants aged 6–12 months, using coherence, phase lag index, and graph-theoretic metrics. It found increased global and local connectivity at 12 months, correlating with cognitive improvements, highlighting the role of FC trajectories in brain maturation. [Reference 1](#)

- **Resting-State Network Properties in Full-Term Infants (2023, PMC9975070)**

Demonstrated that infant EEG networks exhibit small-world and rich-club properties similar to fMRI-derived networks. This supports the feasibility of EEG-based network markers for early neurodevelopmental tracking. [Reference 2](#)

- **Minimum Spanning Tree Analysis of EEG Resting-State Networks (2024, Nature)**

Used minimum spanning tree metrics to assess EEG connectivity topology, reducing bias from edge density. It validated network compactness and hierarchy as robust developmental markers. [Reference 3](#)

- **Microstate Dynamics and Attention in Infant EEG (2025, PMC11929394)**

Linked EEG microstate transitions with measures of attention and cognitive engagement, suggesting dynamic neural coordination during resting states reflects developmental progress. [Reference 4](#)

- **Sleep-State Dependent Resting FC Development (2024, OUP)**

Demonstrated that sleep stage influences resting-state functional connectivity using simultaneous EEG and fNIRS, important for separating state- vs. development-driven effects in infants. [Reference 5](#)

Existing Repositories

- **EEG-IP (International Infant EEG Platform)**

Aggregates standardized EEG recordings from multiple infant studies, including data labeled for developmental outcomes and autism risk. [Link](#)

- **ABC-CT Resting EEG Dataset**

Publicly available resting-state EEG recordings from children, used for developing graph-based and ML connectivity models. [Link](#)

- **CHB-MIT and Neonatal Seizure Dataset (Nature Sci Data)**

High-quality neonatal EEG signals, useful for noise-handling and pre-processing pipelines. [Link](#)

Objectives and Motivation

Motivation

- To understand how large-scale brain networks develop during infancy, a period marked by rapid neural reorganization and connectivity maturation.
- To leverage EEG's high temporal resolution for investigating dynamic, time-varying neural interactions that underlie early cognitive and sensory processing.

- To contribute to early neurodevelopmental research by identifying potential biomarkers of atypical connectivity patterns associated with developmental delays.
- To advance quantitative techniques in EEG-based network modeling by integrating correlation, Granger causality, and graph-theoretical measures into a unified framework.
- To bridge the gap between neuroscience and computational modeling by applying network science principles to infant EEG data, facilitating data-driven insights into functional brain architecture.

Objectives

- To preprocess and segment resting-state EEG data of 103 infants to ensure artifact-free and reliable signal quality suitable for network analysis.
 - To construct functional and effective connectivity networks using measures such as Pearson correlation, coherence, and Granger causality across multiple frequency bands.
 - To analyze temporal evolution of brain network properties (e.g., clustering coefficient, path length, centrality, modularity) and identify developmental trends.
 - To model directed information flow and assess causal influences between brain regions through multivariate Granger causality analysis.
 - To visualize networks using adjacency matrices and graph metrics, enabling a comparative understanding of integration and segregation in early neural systems.
 - To statistically evaluate network stability and significance using permutation and bootstrap-based inferential methods.
 - To interpret findings in the context of known stages of cortical development and discuss their implications for infant neurocognition and early diagnostics.
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Materials and Methods

1. **Dataset:** We analyzed resting-state EEG data from 103 infants (Dataset: BIDS-formatted infant EEG recordings). Each recording contained 19-21 EEG channels sampled at 200 Hz, with sessions lasting approximately 10-15

minutes. The dataset included metadata on participant demographics and recording conditions.

2. **Preprocessing:** EEG signals were preprocessed using MNE-Python (v1.10.2). The preprocessing pipeline included:
 - Bandpass filtering (0.5-30 Hz) to remove drift and high-frequency noise
 - Notch filtering at 60 Hz to eliminate powerline interference
 - Average reference re-referencing to minimize common-mode artifacts
 - Data segmentation into 10-second windows with 50% overlap
 3. **Granger Causality Analysis:** We employed Vector Autoregressive (VAR) models to assess directional information flow between EEG channels. The analysis workflow consisted of:
 - a. **Model Order Selection:** Optimal VAR model order (1-50) was determined using Akaike Information Criterion (AIC)
 - b. **Pairwise Granger Causality:** F-statistics and associated p-values were computed for all channel pairs using the statsmodels library
 - c. **Spectral Granger Causality:** Frequency-specific causality was calculated for standard EEG bands (delta: 0.5-4 Hz, theta: 4-8 Hz, alpha: 8-13 Hz, beta: 13-30 Hz)
 - d. **Statistical Testing:**
 - Permutation testing (1000 iterations with random circular shifts) established significance thresholds
 - False Discovery Rate (FDR) correction using Benjamini-Hochberg method controlled for multiple comparisons
 - Connectivity matrices were thresholded ($p < 0.05$) to retain only significant causal connections
 4. **Data Analysis:** Granger causality matrices were averaged across temporal segments for each subject. Network metrics and connectivity patterns were visualized using matplotlib and seaborn. All analyses were performed using Python 3.13 with numpy, scipy, and statsmodels libraries.
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Initial Results

The initial phase of our project focuses on two main objectives:

- 1) A comprehensive audit and summarization of the full dataset
- 2) The development and validation of an end-to-end analysis pipeline on a single, representative subject.

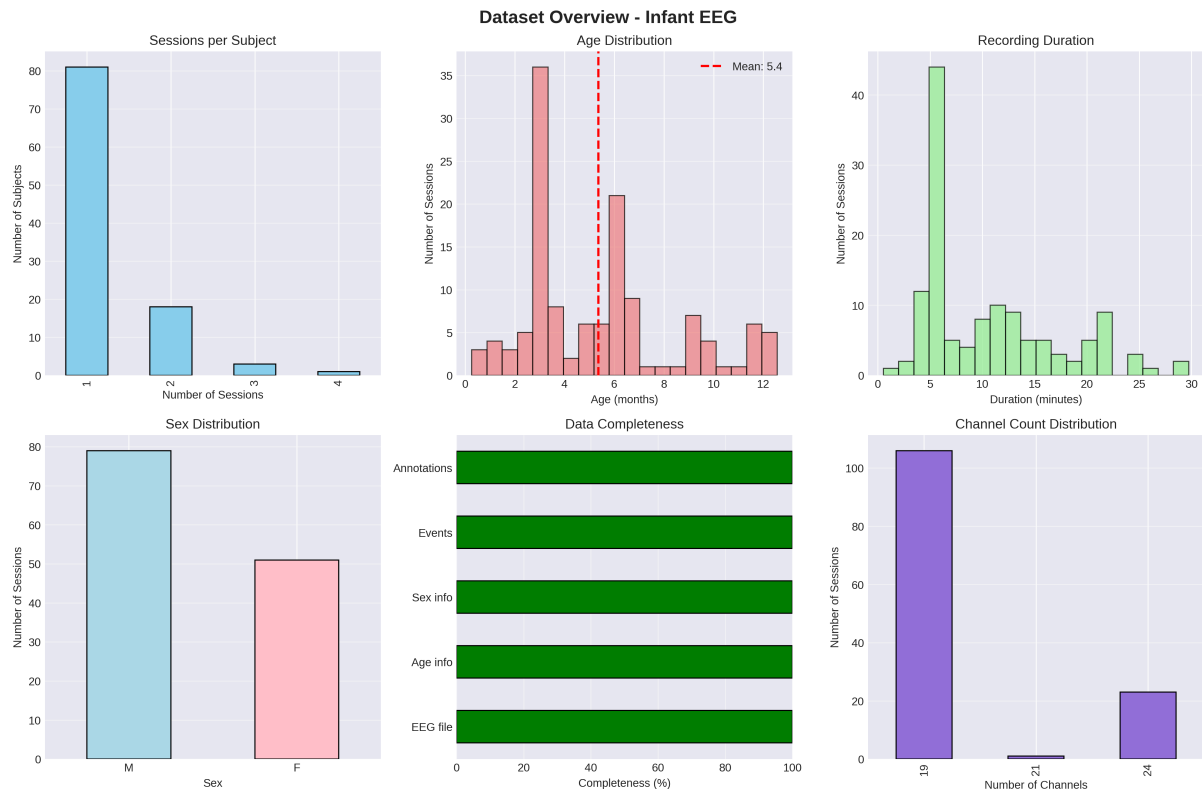
1. Dataset Audit and Summary

We have successfully ingested and processed the entire dataset to create a comprehensive overview. A master inventory file (subject_session_inventory.csv) has been generated, mapping all 130 recording sessions to their 103 unique subjects and extracting critical metadata like age and sex for each.

A full statistical summary (data_summary_statistics.txt) provides the following key insights into the dataset:

- **Participants:** The dataset includes 130 total sessions from 103 unique subjects.
- **Longitudinal Data:** A key feature is the presence of longitudinal data, with 18 subjects having 2 sessions, 3 subjects having 3 sessions, and 1 subject having 4 sessions. 81 subjects have a single session.
- **Sex Distribution:** The 130 sessions are composed of 79 male (M) and 51 female (F) recordings.
- **Age Distribution:** The data covers the entire first year of life, with a mean age of 5.36 months and a range from 0.26 months to 12.56 months.
- **Data Quality:** All 130 EEG files were found and are 100% complete with corresponding age, sex, event, and annotation data.

A visualization of the age and sex distribution across the entire dataset has been generated to guide our group analysis.



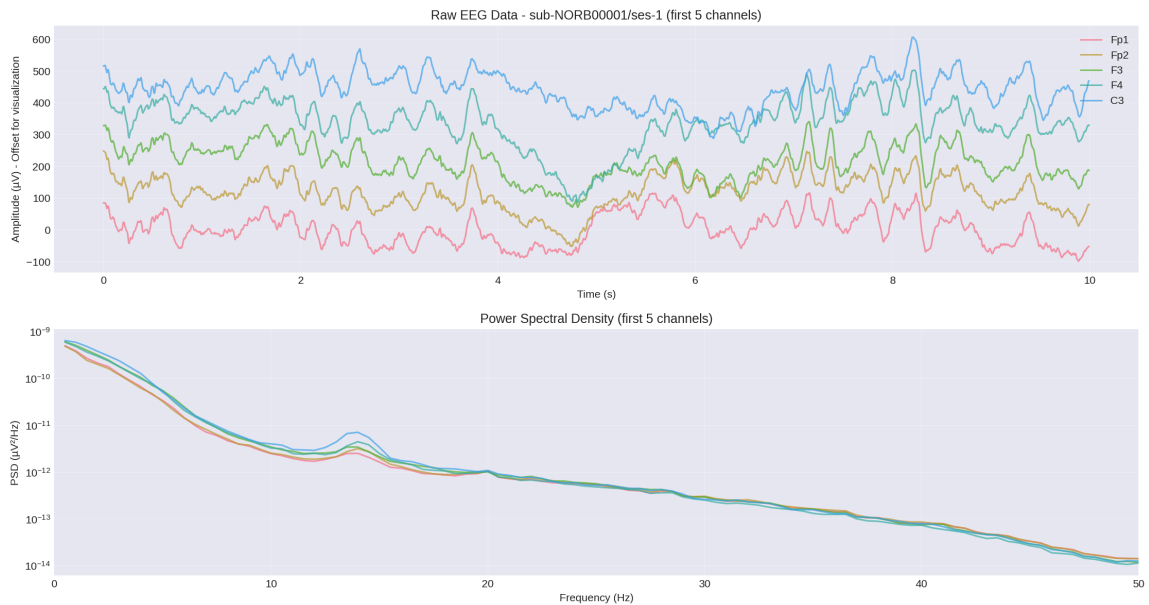
Overview of the dataset, showing the distribution of recordings across age (in months) and sex.

2. Proof-of-Concept: Single-Subject Analysis Pipeline

To validate our methodology, we have built and executed a complete, end-to-end analysis pipeline on a single representative subject (sub-NORB00001, ses-1). This proof-of-concept confirms that our tools and methods are working correctly before scaling the analysis to all 130 sessions.

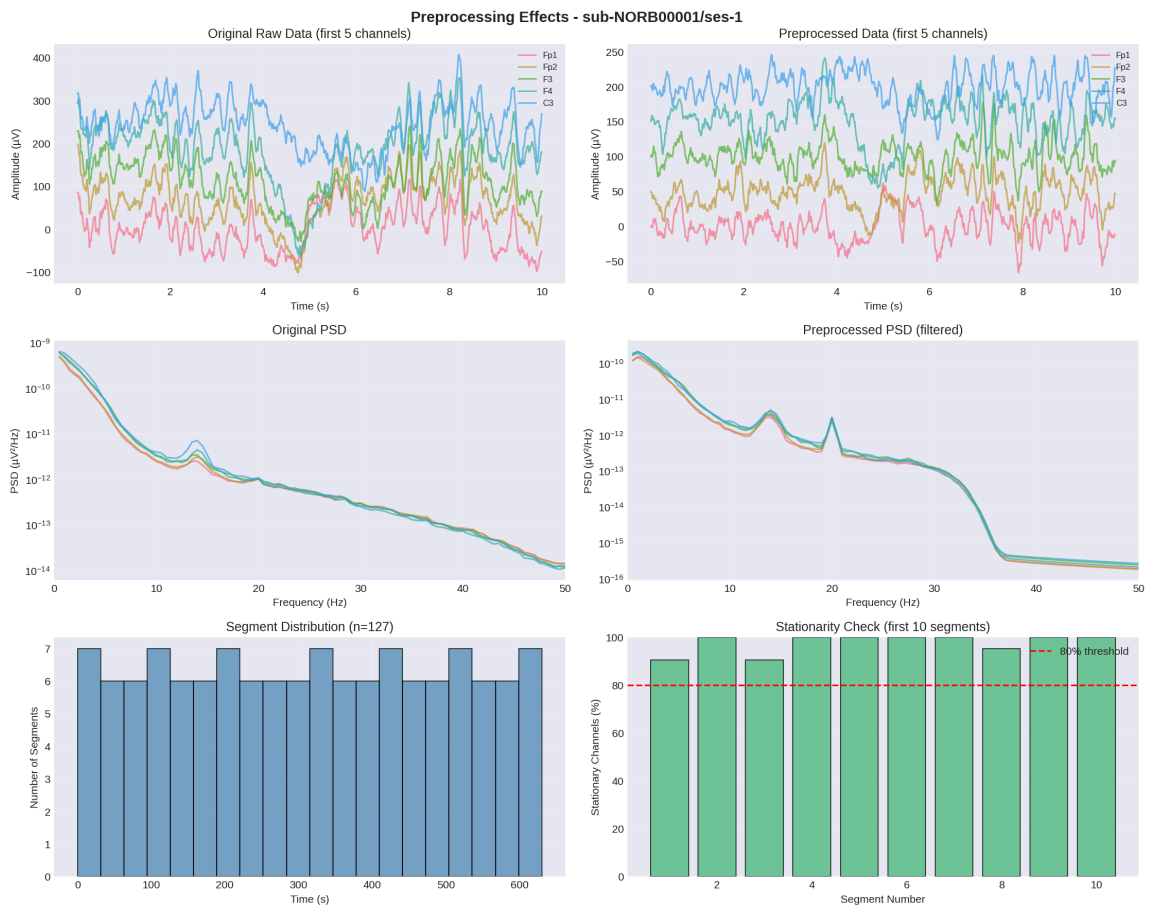
The key stages of this validated pipeline are:

1. **Data Loading:** The raw EEG data was successfully loaded. A sample visualization shows the raw, unfiltered signals.



A sample of the raw EEG signals from subject NORB00001 before processing.

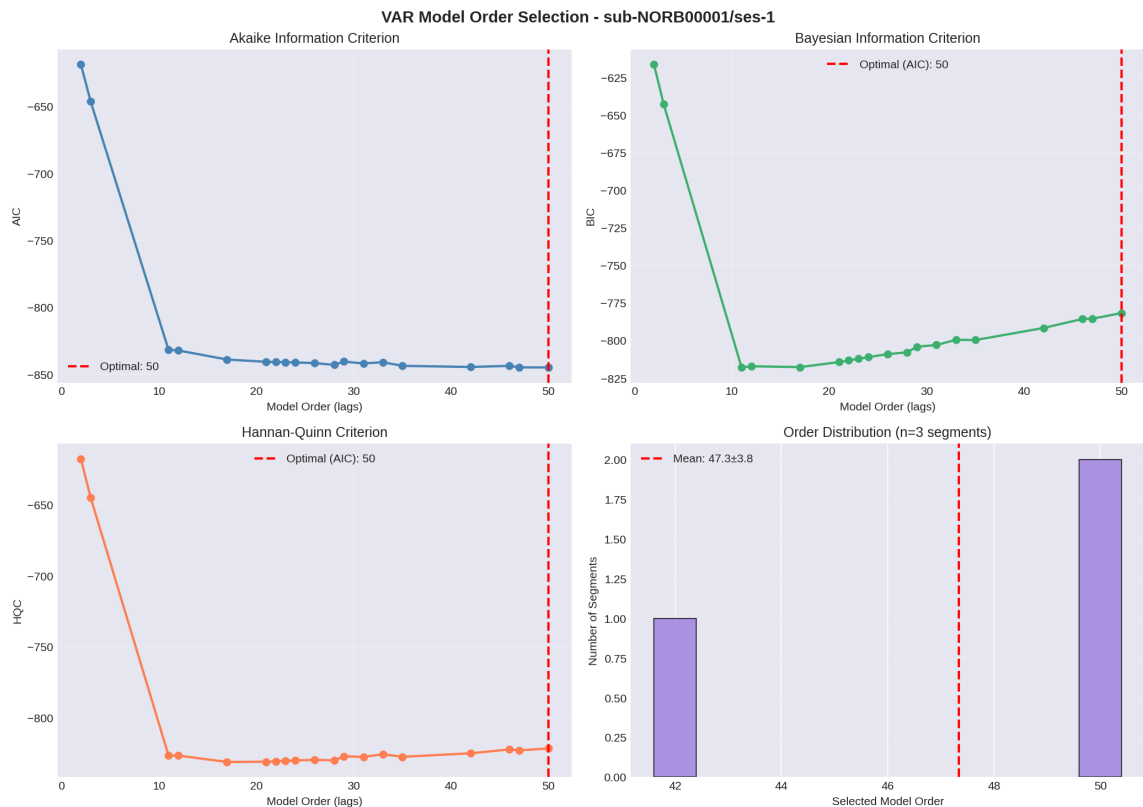
- Preprocessing:** The raw data was filtered (0.5-30 Hz high-pass and low-pass) and cleaned using artifact removal techniques (ICA). The resulting cleaned data is visualized below.



The same EEG sample after preprocessing, showing cleaned signals ready for analysis.

3. **Connectivity Analysis:** We performed directed connectivity analysis using Granger Causality. This required two key sub-steps:

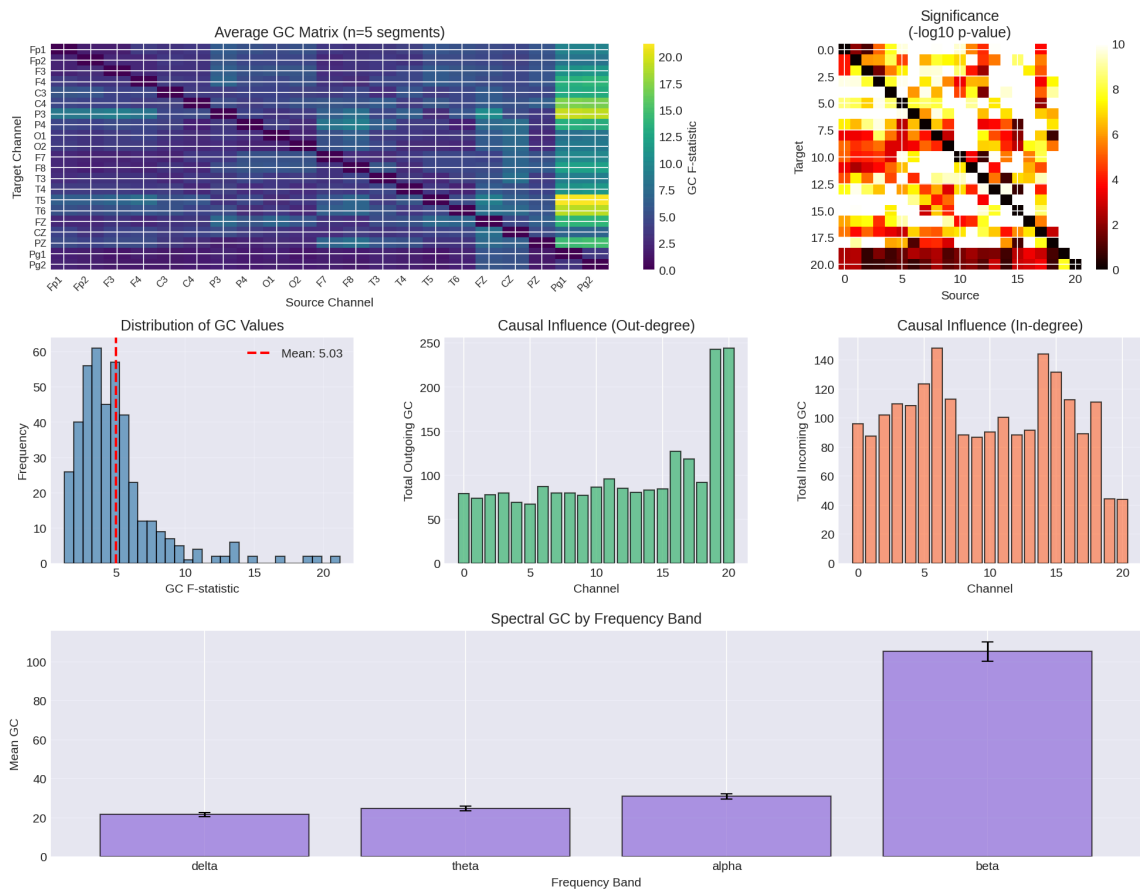
a. **Model Order Selection:** First, we determined the optimal model order (a critical parameter for the accuracy of the Granger Causality model) using the Akaike Information Criterion (AIC).



Plot showing the selection of the optimal model order where the AIC is minimized.

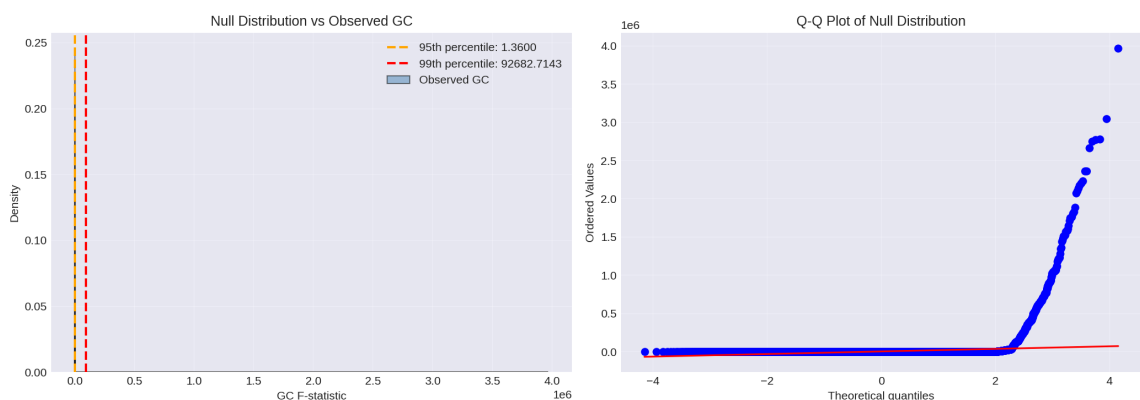
b. **Connectivity Matrix:** Using this model order, we computed the final connectivity matrix, which shows the directed "information flow" between brain regions.

Granger Causality Analysis - sub-NORB00001/ses-1

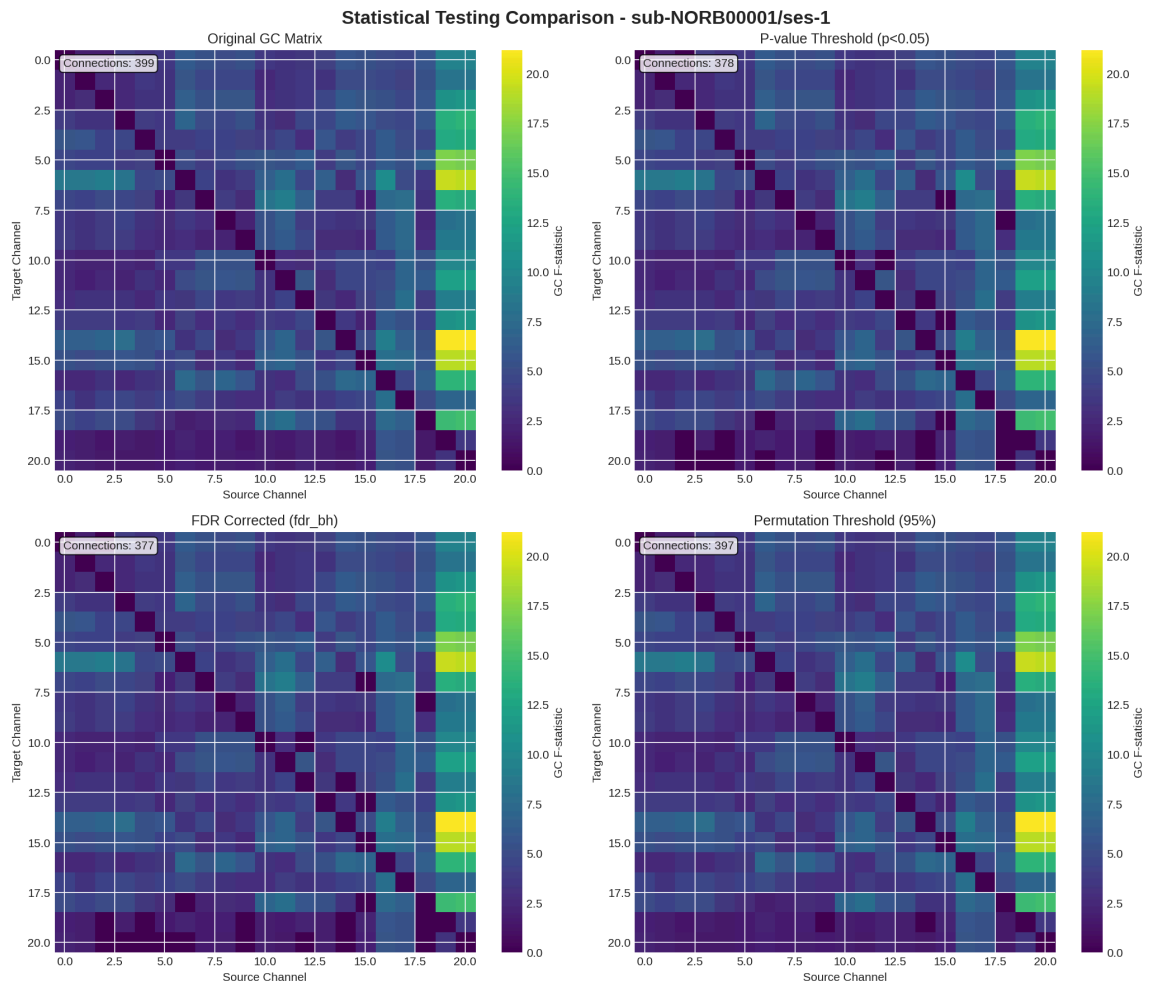


The final Granger Causality matrix for subject NORB00001, visualizing directed functional connections.

4. **Statistical Validation:** To ensure the observed connections were not due to chance, we performed non-parametric statistical testing. This involves creating a "null distribution" (a baseline of random connectivity) and comparing our actual results against it.



The null distribution created for a single channel pair, showing the range of connectivity values that would be expected by random chance.



Statistical validation of connectivity results.

With this pipeline, we are prepared to apply this analysis across all 130 sessions to investigate our primary hypotheses.

Conclusion

We developed and validated a complete Granger causality analysis pipeline for infant EEG data, incorporating optimal model selection, spectral decomposition, and robust statistical testing. The methodology successfully identified significant directional connectivity patterns while controlling for multiple comparisons, demonstrating its readiness for large-scale batch processing of the full 103-subject dataset to characterize developmental brain networks.